

SQL Notes: DISTINCT and EXISTS

What is DISTINCT?

DISTINCT removes duplicate rows from the result.

Suppose a table contains

Student	Department
Aadya	ECE
Arpit	ECE
Riya	CSE

```
SELECT DISTINCT Department  
FROM Students;
```

Result

Department
ECE
CSE

DISTINCT on Multiple Columns

This is often misunderstood.

Suppose

Name	Department
Aadya	ECE
Aadya	ECE
Aadya	IT
Arpit	ECE

Query

```
SELECT DISTINCT Name, Department  
FROM Students;
```

Result

Name	Department
Aadya	ECE
Aadya	IT
Arpit	ECE

Notice

DISTINCT considers the entire row of selected columns, not each column independently.

```
SELECT DISTINCT City, State  
FROM Customers;
```

Even if the city repeats, different states create different combinations.

DISTINCT with NULL

Another favorite interview question.

Suppose

Phone
NULL
NULL
98765
91234

Query

```
SELECT DISTINCT Phone  
FROM Students;
```

Result

Phone
NULL
98765
91234

Although

NULL ≠ NULL

for comparisons,

DISTINCT treats all NULL values as one unique value in the output.

Removing duplicates requires additional work.

The optimizer may use

- Sorting
- Hashing

Large tables with many duplicate values require more processing.

EXISTS

Important Note

EXISTS almost always appears together with subqueries.

What is EXISTS?

EXISTS checks whether a subquery returns at least one row.

It does not care

- what values are returned,
- how many columns are returned,
- or what those values are.

It only asks one question:

Did the subquery return at least one row?

If yes → **TRUE**

Otherwise → **FALSE**

NOT EXISTS

Opposite of EXISTS.

Returns rows where no matching row exists.

Example

Customers who never placed an order.

Performance Advantages

Suppose a customer has 10,000 orders.

EXISTS stops immediately after finding the first matching row.

Found one row?



Stop searching.

Because of this, EXISTS is often more efficient than alternatives when checking existence.

Real-World Examples

Customers with orders

```
SELECT *
FROM Customers c
WHERE EXISTS (
    SELECT 1
    FROM Orders o
    WHERE o.CustomerID = c.CustomerID
);
```

We usually write

```
SELECT 1
```

to make the intention clear:

"I don't care about the data. I only care whether a row exists."

Customers without orders

```
SELECT *
FROM Customers c
WHERE NOT EXISTS (
    SELECT 1
    FROM Orders o
    WHERE o.CustomerID = c.CustomerID
);
```

Employees managing someone

```
SELECT *
FROM Employees e
WHERE EXISTS (
    SELECT 1
    FROM Employees m
    WHERE m.ManagerID = e.EmployeeID
);
```

Sometimes **IN** is simpler.